Multi Agent Automotive Creator

Course Name: Agentic AI

Institution Name: Medi - Caps University – Datagami Skill Based Course

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Project Number: AAI-36

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Academic Year: 2022-26

Problem Statement & Objectives

Problem Statement

The automotive design and development process involves multiple complex stages such as requirement analysis, vehicle configuration, component selection, and design documentation. Traditionally, these tasks are performed manually or using isolated software tools, which leads to increased time consumption, higher costs, and dependency on domain expertise.

Although automation tools exist, they often lack coordination and intelligence to handle multi-step workflows efficiently. With the emergence of **Agentic AI systems**, where multiple intelligent agents collaborate to solve complex problems, there is an opportunity to transform automotive design into a more structured and automated process.

However, there is currently a lack of integrated systems that:

- Utilize a **multi-agent architecture** for task distribution
- Enable seamless communication and coordination between agents
- Automate end-to-end automotive design workflows
- Provide intelligent decision-making using AI models and data

Therefore, the problem addressed in this project is:

To design and implement a Multi-Agent Automotive Creator, an Agentic AI-based system that utilizes multiple intelligent agents coordinated through a centralized orchestration mechanism to automate automotive design processes such as requirement analysis, configuration generation, component selection, and documentation using Python and AI technologies.

This project demonstrates how agent-based systems can improve efficiency, reduce manual effort, and provide scalable solutions for complex automotive workflows.

2. Project Objectives

The main objectives of the Multi-Agent Automotive Creator are:

1. Develop a Multi-Agent Agentic AI System

To build a system consisting of multiple intelligent agents where:

- Each agent performs a specific task in the workflow
- Agents communicate through a centralized orchestrator
- The system is implemented using Python and modular architecture

2. Automate Requirement Analysis

To design an agent responsible for:

- Accepting user inputs related to automotive needs
- Analyzing requirements such as vehicle type, features, and constraints
- Structuring inputs into machine-readable formats for further processing

3. Generate Automotive Configurations

To implement a configuration agent that:

- Converts analyzed requirements into suitable vehicle configurations
- Suggests specifications such as engine type, body style, and features
- Ensures logical consistency in generated configurations

4. Perform Intelligent Component Selection

To develop a component selection agent that:

- Selects appropriate automotive components based on requirements
- Uses predefined knowledge or databases for decision-making
- Ensures compatibility between selected components

5. Generate Design Documentation

To implement a documentation agent that:

- Compiles outputs from all agents
- Generates structured automotive design reports
- Provides clear and professional documentation for presentation

6. Enable Agent Collaboration and Workflow Orchestration

To design a centralized orchestration mechanism that:

- Coordinates communication between agents
- Manages task execution in a sequential or parallel manner
- Ensures smooth workflow completion

7. Build a Scalable and Modular Architecture

To create a system that:

- Is modular and easy to extend
- Supports integration with external tools or databases
- Can be deployed locally or on cloud platforms

3. Scope of the Project

1. Technical Scope

The project includes:

- Design and implementation of multi-agent architecture
- Development of individual agents for specific tasks
- Orchestration logic for agent communication
- Backend development using Python
- Optional integration with databases or external APIs
- User interface for input and output interaction

2. Functional Scope

The system allows users to:

- Input automotive design requirements
- Automatically generate vehicle configurations
- Receive component selection suggestions

Generate complete design documentation

- Modify inputs and regenerate outputs

3. Educational Scope

The system can be used for:

- Academic demonstration of Agentic AI systems

- Final year major project presentations
- Understanding multi-agent collaboration
- Learning workflow automation using AI

4. Industrial Scope

This project has applications in:

- Automotive design and development companies
- Product configuration systems
- Engineering and prototyping teams
- AI-driven automation platforms

5. Future Scope

The project can be extended to:

- Integration with real-time automotive databases
- Advanced decision-making using machine learning models
- 3D vehicle modeling and simulation
- Integration with CAD tools
- Deployment as a SaaS-based platform

6. Limitations

- Limited by predefined rules or knowledge base
- Requires proper orchestration for agent coordination
- May not fully replicate expert-level engineering decisions
- Dependent on system design and data quality
- Does not perform real-world mechanical validation

Proposed Solution

1. Key Features

- Multi-Agent Architecture for task distribution
- Centralized Orchestration Mechanism
- Automated Requirement Analysis
- Intelligent Vehicle Configuration Generation
- Component Selection System
- Automated Design Documentation
- Modular and Scalable System Design
- User Input-Based Customization
- Workflow Automation across multiple agents

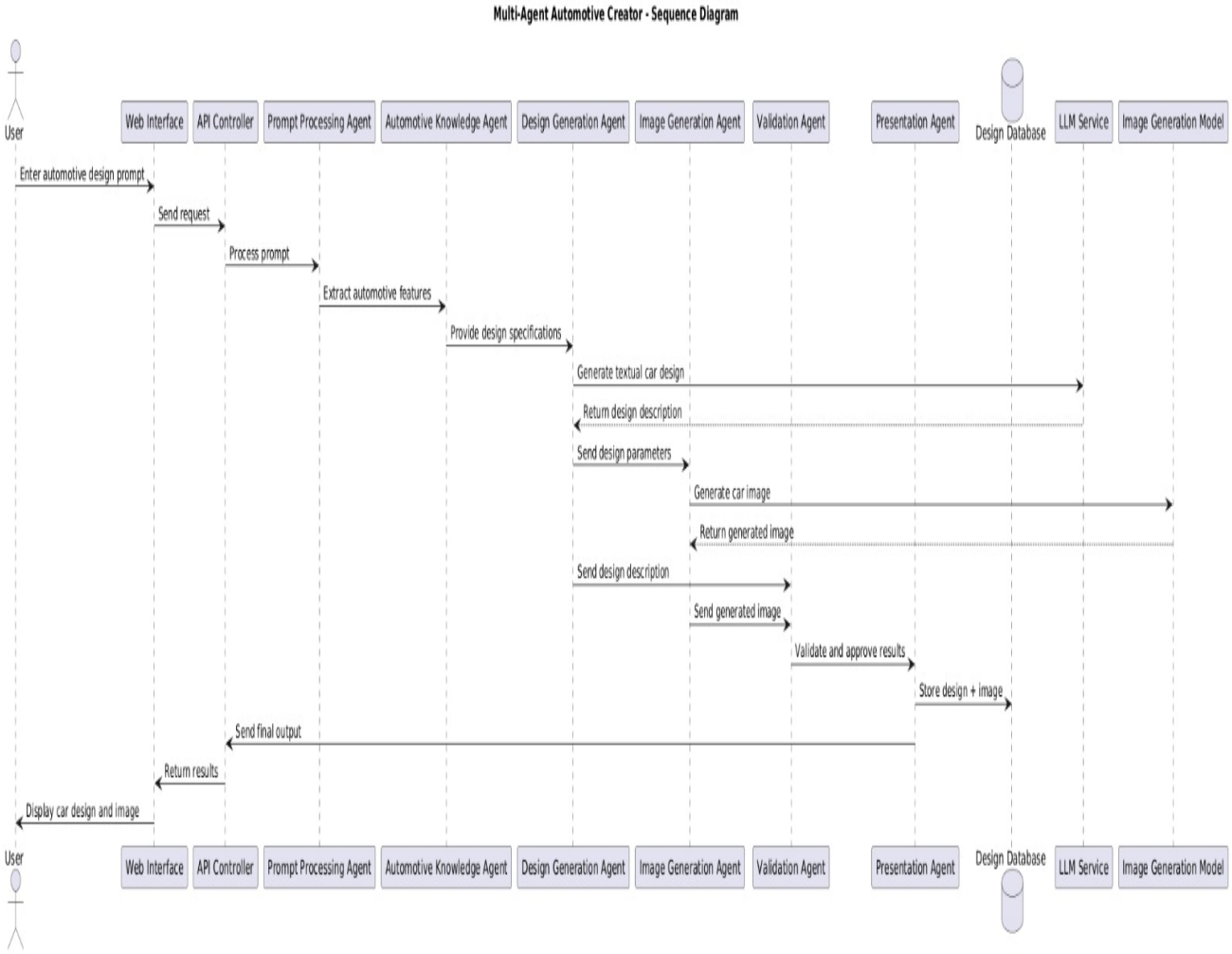
  Integration capability with external tools and databases

2. Overall Architecture / Workflow

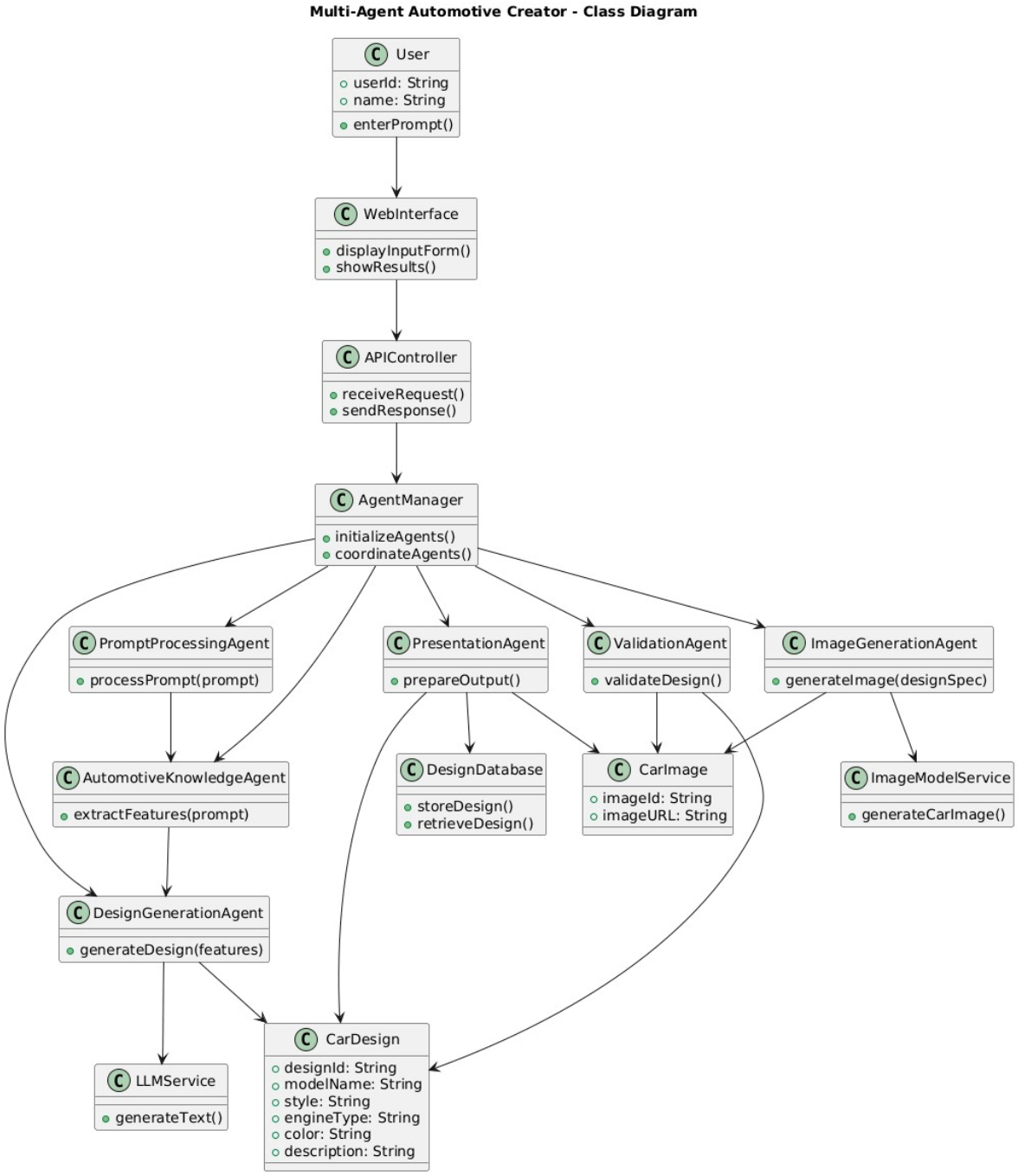
The **Multi-Agent Automotive Creator** follows an agent-based workflow where multiple intelligent agents collaborate through a centralized orchestration system to complete the automotive design process efficiently.

Step-by-Step Workflow:

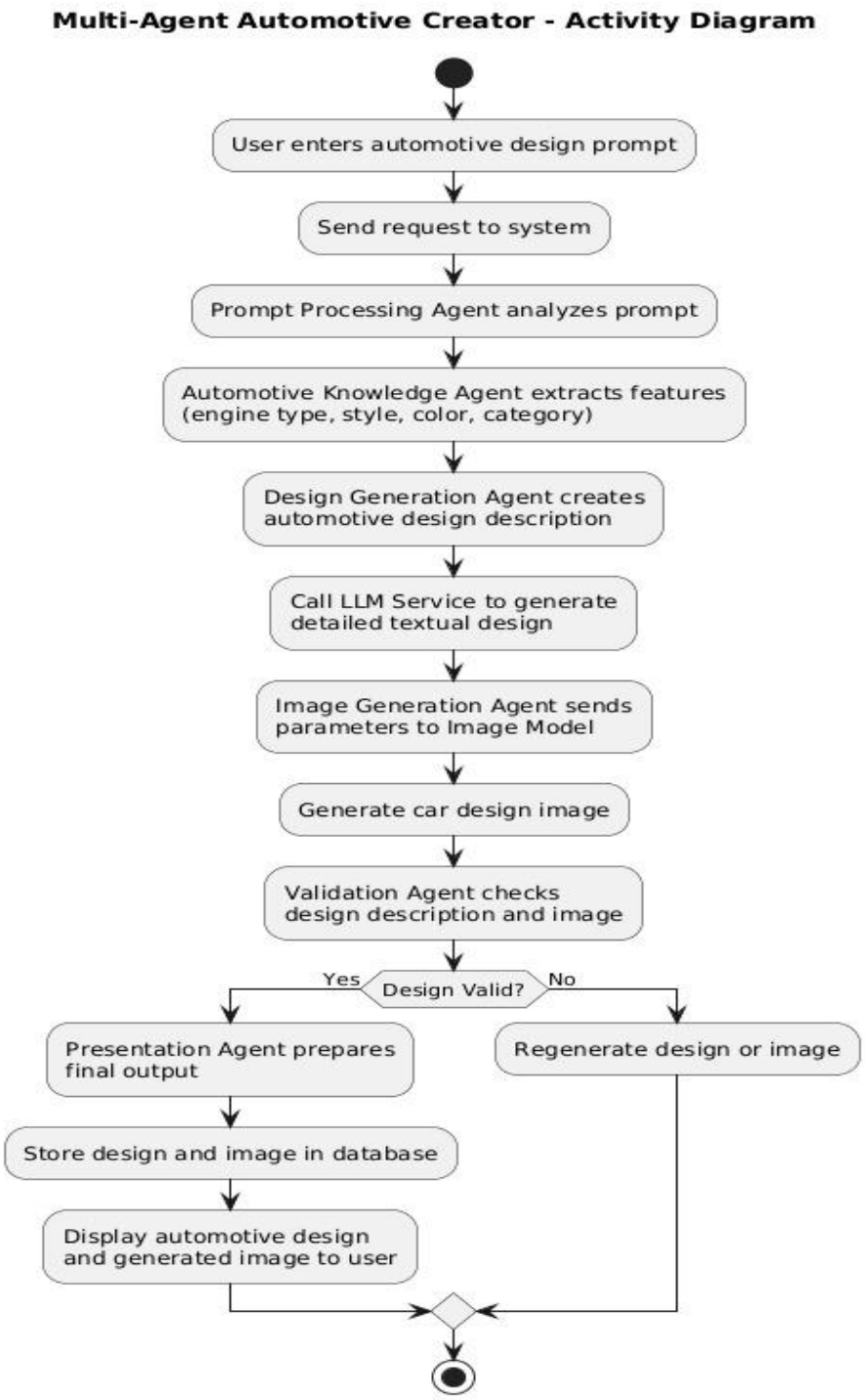
1.Sequence Diagram :-



2.Class diagram:-



3. Activity diagram:-



1. **User Input Stage**
    - The user provides automotive design requirements such as vehicle type, features, and constraints through the interface.
2. **Requirement Analysis Agent**
    - This agent processes and interprets user inputs.
    - Converts raw input into structured data for further processing.
3. **Orchestrator (Central Controller)**
    - Manages communication between all agents.
    - Decides the sequence of task execution.
    - Ensures smooth workflow coordination.
4. **Configuration Agent**
    - Generates suitable vehicle configurations based on analyzed requirements.
    - Defines specifications such as engine type, body style, and features.
5. **Component Selection Agent**
    - Selects appropriate components based on configuration.
    - Ensures compatibility and logical consistency.
6. **Knowledge/Data Support (Optional Integration)**
    - Agents may retrieve relevant data from databases or external sources to improve accuracy.
7. **Documentation Agent**
    - Compiles outputs from all agents.
    - Generates structured automotive design documentation.
8. **Output Generation**
    - Final results are presented to the user in a readable format.
    - Users can modify inputs and regenerate results if needed.

Architecture Summary

Input Layer: User Interface

- **Processing Layer:** Multiple Intelligent Agents
- **Control Layer:** Central Orchestrator
- **Output Layer:** Generated Configurations + Documentation

3. Tools & Technologies Used

Programming Language

- Python

Frameworks / Libraries

- LangChain / CrewAI (for multi-agent workflow, if used)
- FastAPI / Flask (for backend development)

AI / Agentic Components

- Large Language Models (LLMs) for reasoning and generation
- Rule-based or AI-based decision logic for agents

Database / Storage

SQLite / MongoDB (for storing configurations and data)

Frontend (Optional)

- HTML, CSS, JavaScript

  Streamlit (for quick UI development)

Development Tools

- VS Code
- Git & GitHub

Integration Capabilities

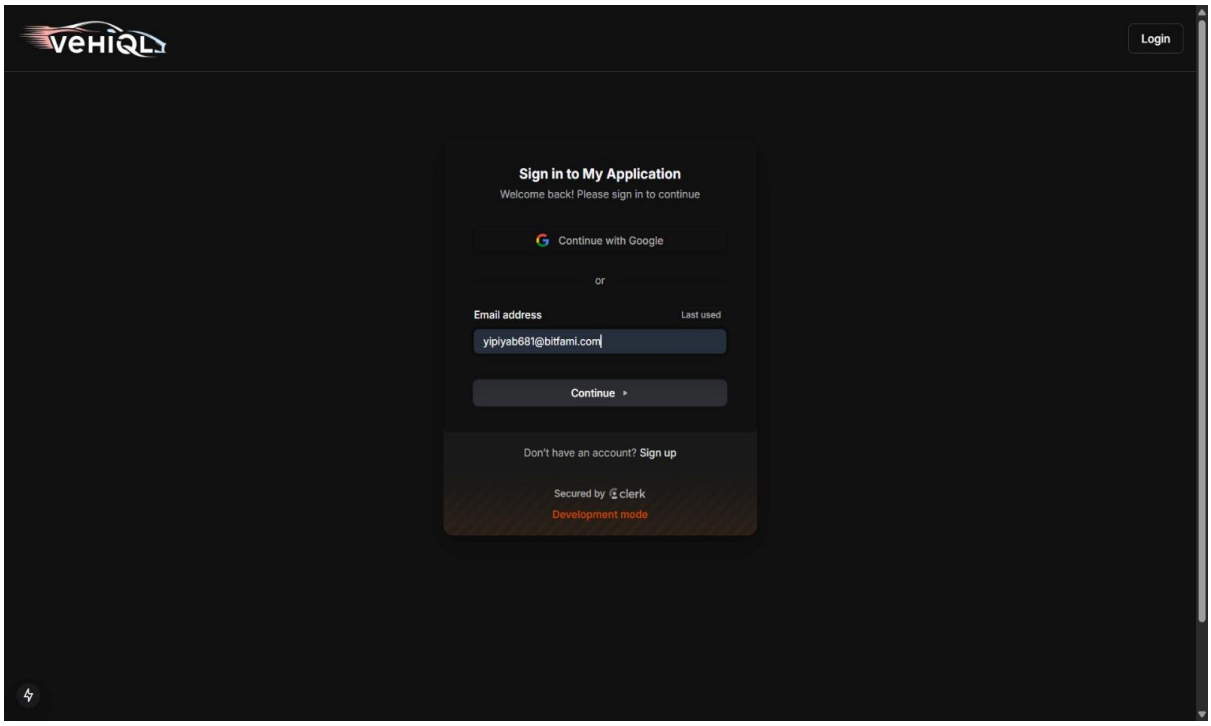
- External APIs (if required)
- Knowledge bases or datasets

Results & Output

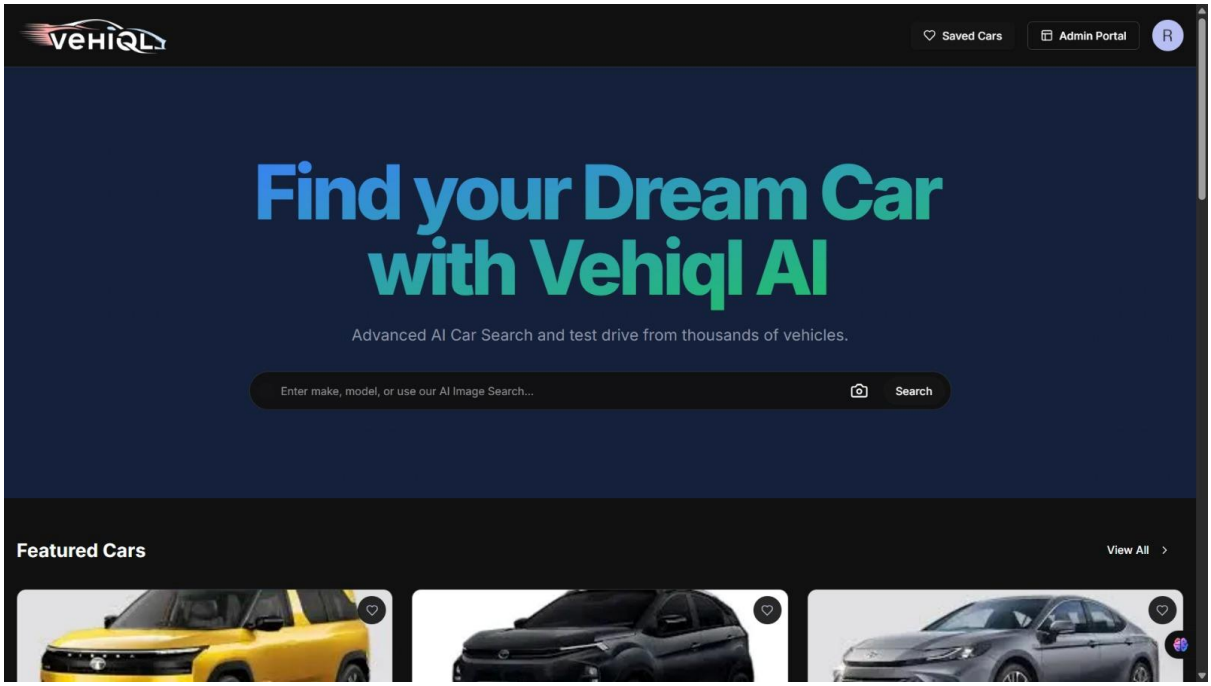
The **Multi-Agent Automotive Creator system** was successfully implemented and tested to validate its agent-based workflow. The system demonstrates how multiple intelligent agents collaborate through a centralized orchestration mechanism to automate the automotive design and development process from user requirements to final documentation.

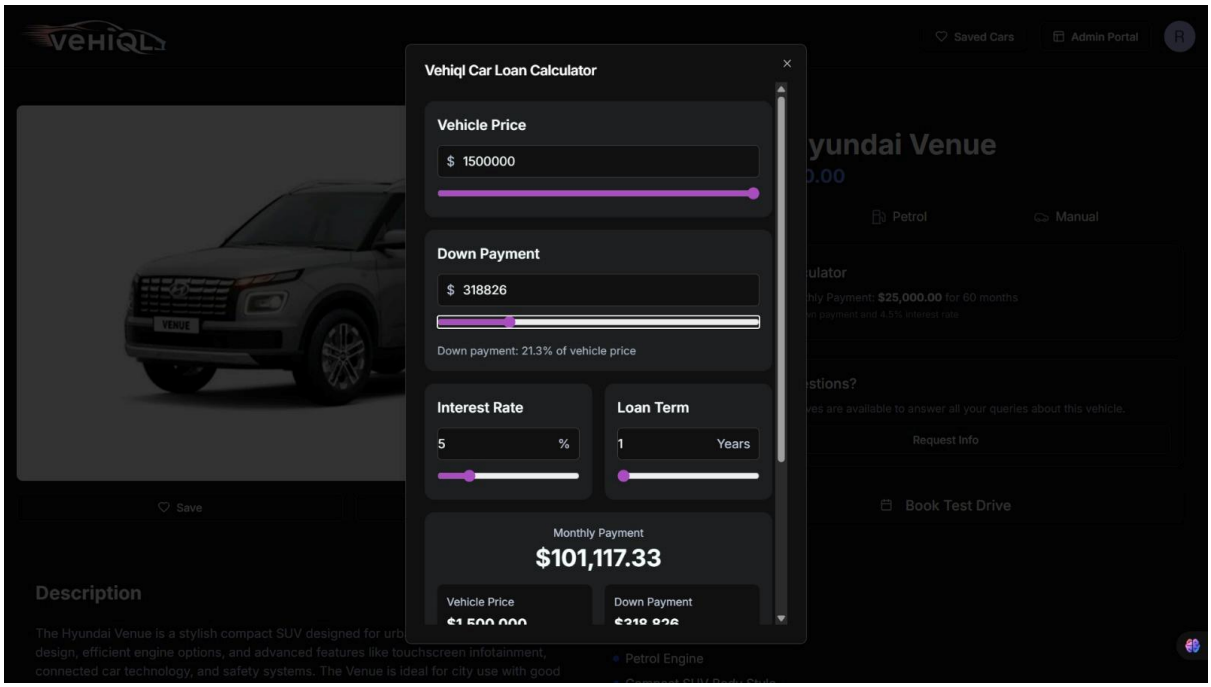
1. Screenshots / Outputs

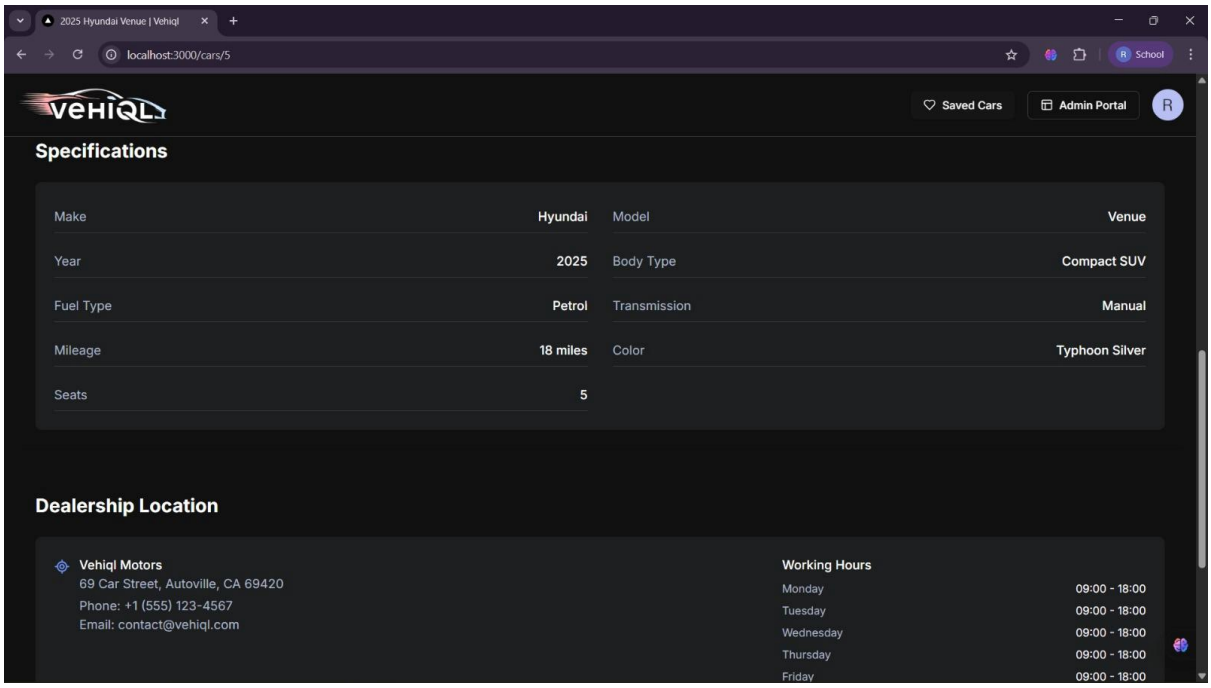
1.Login page

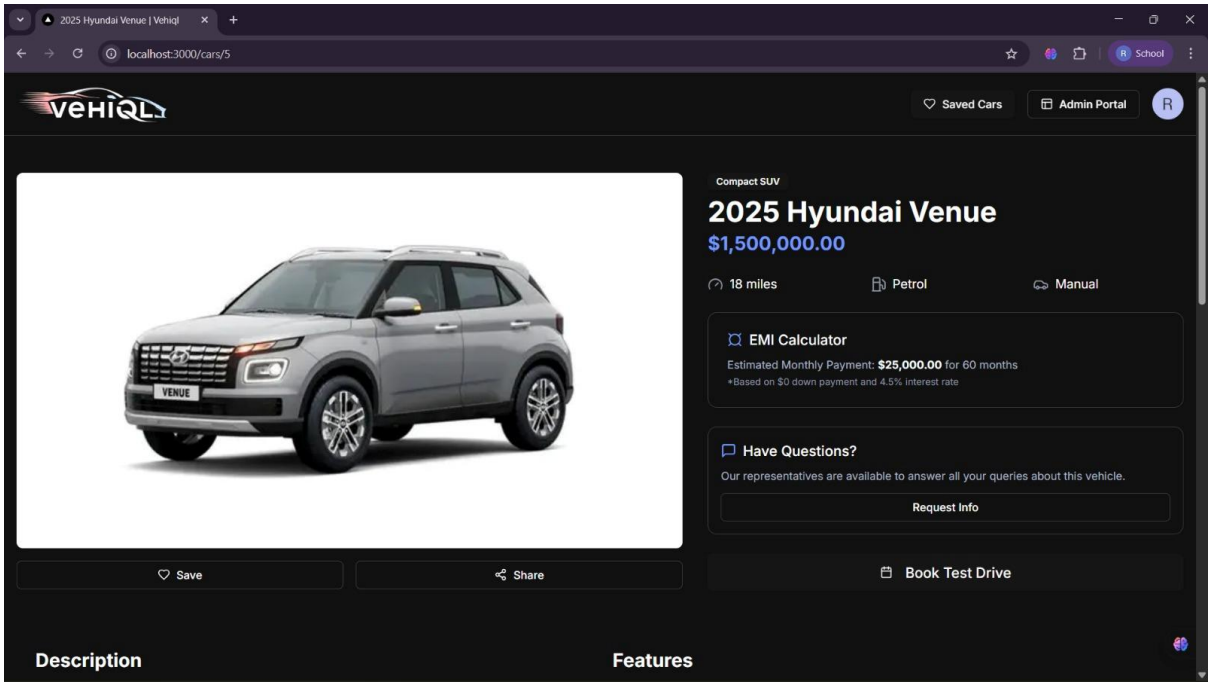


1.Admin portal

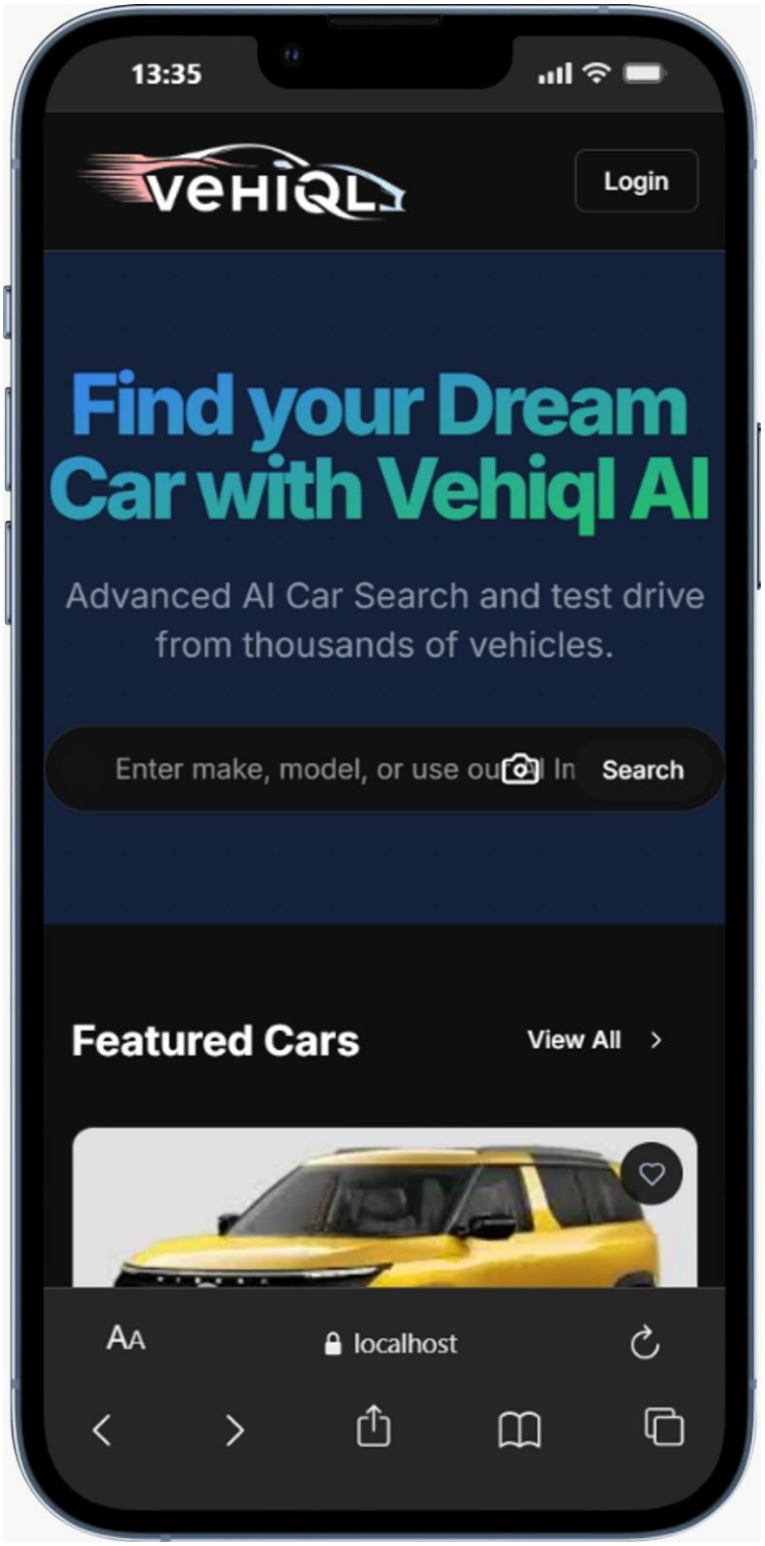





2025 Hyundai Venue | Vehiql
localhost:3000/cars/5
VEHIQL
Saved Cars
Admin Portal
Specifications
Make Hyundai
Model Venue
Year 2025
Body Type Compact SUV
Fuel Type Petrol
Transmission Manual
Mileage 18 miles
Color Typhoon Silver
Seats 5
Dealership Location
Vehiql Motors
69 Car Street, Autoville, CA 69420
Phone: +1 (555) 123-4567
Email: contact@vehiql.com
Working Hours
Monday 09:00 - 18:00
Tuesday 09:00 - 18:00
Wednesday 09:00 - 18:00
Thursday 09:00 - 18:00
Friday 09:00 - 18:00


2025 Hyundai Venue | Vehiql
localhost:3000/cars/5
VEHIQL
Saved Cars
Admin Portal
Compact SUV
2025 Hyundai Venue
$1,500,000.00
18 miles
Petrol
Manual
EMI Calculator
Estimated Monthly Payment: $25,000.00 for 60 months
*Based on $0 down payment and 4.5% interest rate
Have Questions?
Our representatives are available to answer all your queries about this vehicle.
Request Info
Save
Share
Book Test Drive
Description
Features

3. Responsive



a: User Input Interface

- Web-based interface developed using HTML/CSS or Streamlit.
- Input fields for automotive design requirements (vehicle type, features, constraints).
- "Generate Configuration" button to initiate the workflow.

b: Processed Requirement Output (Requirement Analysis Agent)

The Requirement Analysis Agent successfully processed user input and generated structured data including:

- Vehicle Type (e.g., SUV, Sedan, Electric Vehicle)
- Feature Requirements (e.g., autonomous driving, infotainment system)

  Performance Constraints (e.g., range, fuel type, efficiency)
- User Preferences and design priorities

This output confirms accurate interpretation and structuring of raw user input.

Example Input:
"Electric SUV with long range, advanced safety features, and modern interior design suitable for urban and highway driving."

c: Generated Automotive Configuration (Configuration Agent)

The Configuration Agent produced a detailed vehicle configuration including:

- Body Type: SUV
- Powertrain: Electric motor system
- Battery Capacity: Long-range battery pack
- Key Features: Advanced driver assistance systems, smart connectivity
- Design Aspects: Modern styling with aerodynamic structure

This demonstrates effective transformation of requirements into logical automotive configurations.

d: Component Selection Output (Component Selection Agent)

The Component Selection Agent generated compatible component recommendations such as:

- Battery system and power modules

  Safety components and sensors
- Infotainment and control systems
- Structural and material components

The output ensured compatibility and consistency across selected components.

e: Generated Design Documentation (Documentation Agent)

The Documentation Agent compiled outputs from all agents into a structured report including:

- Requirement Summary
- Vehicle Configuration Details
- Component Selection Information
- Final Design Overview

This confirms successful automation of documentation generation.

2. Reports / Dashboards / Models

A. System Workflow Model

The implemented workflow follows a multi-agent pipeline:

User Input
→ Requirement Analysis Agent
→ Orchestrator
→ Configuration Agent
→ Component Selection Agent
→ Documentation Agent
→ Output Display

This model demonstrates effective collaboration between agents and structured workflow execution.

B. Architecture Model

The system architecture consists of:

1. **Presentation Layer (Frontend UI)**
   - Handles user interaction and input
2. **Application Layer (Python Backend)**
   - Implements business logic and agent coordination
3. **Agent Layer (Multi-Agent System)**
   - Requirement Analysis Agent
   - Configuration Agent
   - Component Selection Agent
   - Documentation Agent
4. **Data Layer (Optional Database / Knowledge Base)**
   - Stores automotive data and configurations

This layered architecture ensures modularity, flexibility, and scalability.

C. Performance Observation Report

During system testing:

- Requirement processing time: 1–3 seconds
- Configuration generation time: 2–4 seconds
- Component selection time: 2–5 seconds
- Documentation generation time: 1–2 seconds

Additional observations:

- Smooth coordination between agents via orchestrator
- Accurate and consistent outputs across multiple runs
- Efficient handling of repeated and modified user inputs

No major system failures during execution

The system maintained stability and responsiveness throughout testing.

3. Key Outcomes

The major outcomes of this project include:

- Successful implementation of a multi-agent architecture for automotive design automation
- Effective coordination between multiple intelligent agents using an orchestration mechanism
- Automation of requirement analysis, configuration generation, and documentation
- Generation of structured and logically consistent automotive outputs
- Development of a modular Python-based system architecture
- Demonstration of Agentic AI workflow in a real-world automotive domain
- Reduction in manual effort and time required for automotive design processes
- Creation of an academic and practical demonstration tool for agent-based AI systems

Final Outcome Statement

The project demonstrates that utilizing a **multi-agent (Agentic AI) approach** for automotive design enables efficient task distribution, intelligent decision-making, and automated workflow execution. By coordinating multiple specialized agents within a unified system, it is possible to generate structured automotive configurations and documentation in a scalable and efficient manner, making the system suitable for academic demonstrations, prototyping environments, and future AI-driven automation applications.

Conclusion

The Multimodal Automotive Creator using Agentic AI project successfully demonstrates the integration of multiple intelligent agents to generate both textual and visual automotive concepts from user input. By combining natural language processing with generative AI techniques, the system effectively transforms simple prompts into detailed vehicle descriptions along with corresponding design visuals.

During the development process, we implemented an agent-based architecture where different agents collaboratively perform tasks such as prompt analysis, content generation,

and image synthesis. This modular design enhances the system's efficiency, scalability, and flexibility. The project also emphasized the importance of seamless frontend-backend integration and user-friendly interface design.

Overall, this project highlights the growing potential of Agentic AI in creative and industrial domains like automotive design. It demonstrates how intelligent systems can automate idea generation and support innovation. The project also provides a strong foundation for future enhancements such as real-time customization, 3D model generation, and advanced simulation capabilities.

Key Learnings

- Understanding of **Agentic AI architecture** and multi-agent collaboration
- Practical experience with **Large Language Models (LLMs)** and prompt engineering
- Knowledge of **multimodal AI systems** (text + image generation)
- Hands-on experience in **API integration and backend development**
- Improved skills in **frontend design using HTML/CSS for user interaction**
- Understanding of **workflow automation between different AI agents**

- Experience in **problem-solving and debugging real-world AI systems**
- Insight into **scalability and modular system design**
- Exposure to **real-world applications of Generative AI in automotive domain.**

Future Scope & Enhancements

a. Future Scope

3D Model Generation
Future system can create complete 3D car models instead of only 2D images

- **Real-Time Customization**
  Users will be able to modify car designs instantly (color, shape, features).

- **Industry Application**
  Can be used by automotive companies for fast concept design and prototyping.
- **Cloud Deployment**
  Making the system online will allow access from anywhere and better scalability.
- **Educational Use**
  Helpful for students to learn vehicle design and AI-based creativity tools.

b. Enhancements

- **More Intelligent Agents**
  Add agents for performance analysis, cost estimation, and material selection.
- **Better UI/UX Design**
  Improve interface to make it more interactive and user-friendly.
- **High-Quality Visual Output**
  Use advanced models to generate more realistic and detailed car designs.
- **Voice Input Feature**
  Allow users to give commands using voice instead of typing.
- **Feedback Learning System**
  System can improve by learning from user choices and preferences.
- **Faster Processing**
  Optimize system to reduce response time and improve performance.
- **Multi-Language Support**
  Enable different languages for wider user accessibility.